

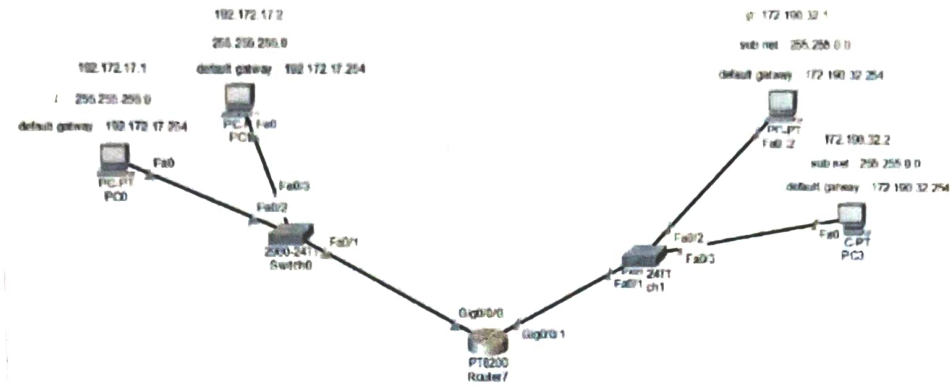
# Implementation of subnetting in CISCO PACKET TRACER simulator.

Aim: To implement the subnetting in CISCO Packet Tracer simulator.

## Procedure:

- (i) open Cisco Packet tracer and create new network topology.
- (ii) Add the required routers, switches and PCs to the workspace.
- (iii) Connect the device using appropriate Copper cables.
- (iv) use the network  $192.168.1.0/24$  and divide it into 27 networks.
- (v) Configure the router interfaces with the required IP addresses and subnet mask  $255.255.255.224$ .
- (vi) use enable and Configure terminal in the router CLI.
- (vii) Enter the interface Configuration terminal in the router and assign the IP using (i) IP address  $\langle \text{IP address} \rangle 255.255.255.224$ .  
(ii) No shutdown.
- (viii) Configure the required switch ports in access mode.
- (ix) Assign each PC its IP address, subnet mask and default gateway.
- (x) Repeat the Configuration for all required subnets.
- (xi) use the ping command to test Connectivity between PCs and router.

(Xii) If Ping is successful, the network is Configured Correctly.



```

Router>
*Sep 24, 11:17:27.1717: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
Router>
Router>
Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#

```

Result:

Thus the implementation of subnetting in Cisco packet tracer simulator is verified.